

LAB 2 [Group # → 1 to 8]

[Platform → HACKERRANK]

Introduction To Programming

(21-01-22)

SOLUTIONS

Q1 → TRIBONACCI SEQUENCE

```
#Q1 Tribonacci Sequence
n = int(input())
a1 = 0
a2 = 1
a3 = 1

if (n==1):
    print(a1)
elif(n==2):
    print(a2)
elif (n==3):
    print(a3)
else:
    ## none of the base cases, then proceeding further
    for i in range(4, n+1):
        tribo_sum = a1+a2+a3
        a1=a2
        a2=a3
        a3=tribo_sum

    print(tribo_sum)
```

Q2 → SMALLEST 2

```
#Q2 smallest 2

small_1 = 100001
small_2 = 100002
n = int(input())
## input taken
for i in range(n) :
    x = int(input())
    if x < small_1 :
        small_2 = small_1
        small_1 = x
    elif x < small_2 :
        small_2 = x
print(f'{small_1} {small_2}')
```

Q3 → FRESHERS PARTY

```
#Q3 Freshers Party

x1 = int(input())
x2 = int(input())
x3 = int(input())
x4 = int(input())
## 4 input has been taken

prod = x1
## variable prod[stores product] has been initialized to x1
if not(x1==x2):
    prod = prod * x2
if not(x3==x2) and not(x3==x1):
    prod = prod * x3
if not(x4==x1) and not(x4==x2) and not(x4==x3):
    prod = prod * x4

print(prod)
##printing the product
```

Q4 → BABYLONIAN ALGORITHM

```
#Q4 babylonian algorithm

n = int(input())
guess_prev = float(input())
##
guess_next = ((n/guess_prev) + guess_prev)/2
while (guess_next- guess_prev) >0.0001:
    ## steps done as mentioned in the problem
    guess_prev = guess_next
    guess_next = ((n/guess_prev) + guess_prev)/2

print(f'{guess_next:.2f}')
## formatted output --> Remember you were not asked
## 'round' the solution
```

Q5 → LEAP YEAR

```
#Q5 Leap year

year = int(input())

if year%400 == 0:
    print("YES")
elif year%100 == 0:
    print("NO")
elif year%4 == 0:
    print("YES")
else:
    print("NO")
```

Q6 → FIND DIVISORS

```
#Q6 Find Divisors
n = int(input())
## input has been taken
## Now loop would be run to find all
## the successive divisors

for num in range(1,n+1):
    ##Why n+1? -> because 'range(1,n)' runs from 1 to (n-1)
    if (n%num) == 0:
        ## that is num divides n
        print(num)
```

Q7 → WHAT'S YOUR BONUS

```
#Q7 What's Your bonus

service_year = int(input())
## number of years, one has worked
## in the company
salary = int(input())
## salary of the employee

## based on the logistics mentioned
if service_year > 8:
    bonus=15/100*salary

elif service_year >=4 and service_year <=8:
    bonus = 10/100*salary

else:
    ## less than 4 years of experience
    bonus = 5/100*salary

print(int(bonus))
```